

MAKTAB DAVOMAD

Face ID Nazorat Tizimi

TEXNIK TOPSHIRIQ

Hujjat nomi:	Maktab Davomad Face ID Tizimi - TZ
Versiya:	1.0
Sana:	14.03.2026
Platforma:	Web + Telegram Bot + Mobile
Arxitektura:	Yagona markazlashgan server
Til:	O'zbek tili

Bu hujjat Maktab Davomad Face ID tizimining texnik topshirig'i bo'lib, tizimning barcha komponentlari, arxitekturasi, API endpointlari, ma'lumotlar bazasi sxemasi va integratsiya talablarini batafsil belgilab beradi.

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1. LOYIHA HAQIDA UMUMIY MA'LUMOT

1.1 Maqsad va vazifalar

Maktab Davomad Face ID Tizimi — maktab o'quvchilarining davomatini Face ID qurilmalar orqali avtomatik ravishda qayd etuvchi, ota-onalarga real vaqtda xabarnoma yuboruvchi va maktab rahbariyatiga to'liq tahliliy ma'lumot taqdim etuvchi yagona markazlashgan platforma.

Asosiy vazifalar:

- Face ID qurilmalar orqali o'quvchilarning kelishi va ketishini avtomatik qayd etish
- Ota-onalarga Telegram bot orqali real vaqtda xabarnoma yuborish
- Maktab rahbariyatiga kunlik, haftalik va oylik hisobotlar taqdim etish
- Barcha maktablarni yagona markazlashgan serverga ulash
- Ko'p bosqichli rol-asosli kirish nazoratini ta'minlash
- Face ID qurilmalar bilan ikki tomonlama integratsiya (push/pull)

1.2 Asosiy prinsiplar

Markazlashgan arxitektura	Barcha maktablar bitta markaziy serverga ulangan. Lokal server yo'q.
Real-time	WebSocket orqali tezkor yangilanishlar. Xabarlar 1-3 soniya ichida yetkaziladi.
Ikki tomonlama integratsiya	Qurilmaga ma'lumot yuklash va qurilmadan log olish imkoniyati.
Modul arxitektura	Har bir komponent mustaqil. Yangi qurilma brendini osongina qo'shish mumkin.
Ko'p rol qo'llab-quvvatlash	Bir foydalanuvchi bir vaqtda bir nechta rolga ega bo'lishi mumkin.

2. TEXNIK STEK

2.1 Backend

Framework	Django 4.2 + Django REST Framework
Ma'lumotlar bazasi	PostgreSQL 15
Cache va Navbat	Redis 7 + Celery 5
WebSocket	Django Channels 4 + Daphne
Autentifikatsiya	JWT (SimpleJWT)
API Hujjati	drf-spectacular (Swagger/Redoc)
Konteyner	Docker + docker-compose

2.2 Frontend

Framework	React 19 + Vite 8
Stilizatsiya	Tailwind CSS 4
State management	Redux Toolkit
Routing	React Router DOM 7
HTTP klient	Axios
Grafiklar	Recharts
Formlar	React Hook Form + Zod

2.3 Telegram Botlar

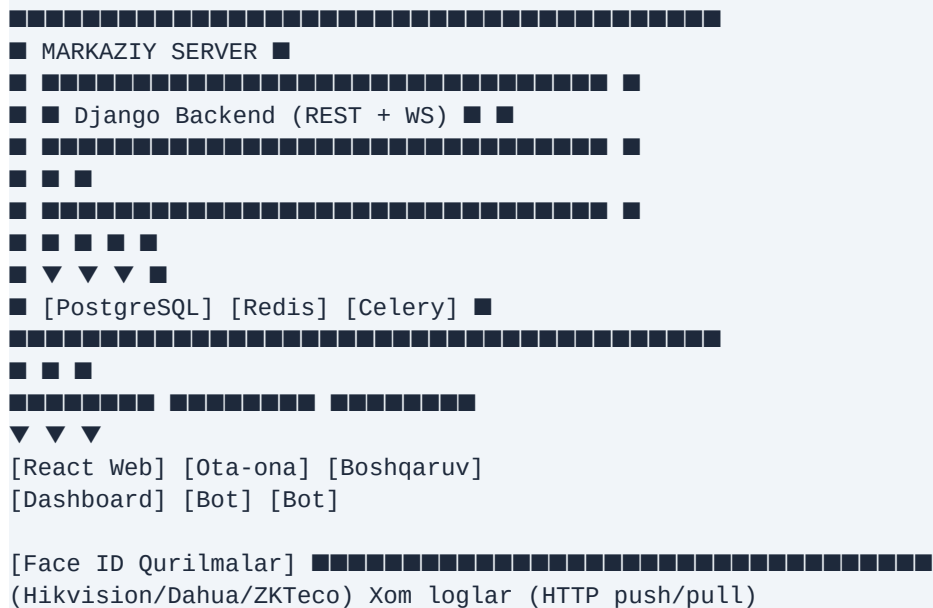
Kutubxona	aiogram 3.x (Python)
Ota-ona boti	Real-time xabarnomalar + davomad ko'rish
Boshqaruv boti	Hisobotlar + qurilma holati + ogohlantirishlar

2.4 Qurilma Integratsiyasi

Hikvision	ISAPI REST protokoli (HTTP Digest auth)
Dahua	HTTP API + CGI protokoli
ZKTeco	HTTP/TCP protokoli
Arxitektura	Adapter pattern - yangi brendlarni osongina qo'shish

3. TIZIM ARXITEKTURASI

3.1 Umumiy arxitektura sxemasi



3.2 Ma'lumotlar oqimi

#	Bosqich	Tavsif
1	Face ID qurilma	O'quvchi yuzini tanib, event log yaratadi
2	Raw log	Qurilma raw logni markaziy serverga yuboradi yoki server pull qiladi
3	Deduplication	Server DeviceRawLog jadvalida takrorlanishni tekshiradi
4	Student mapping	employee_no → student_id orqali o'quvchi aniqlanadi
5	Attendance engine	Check-in/check-out, kechikish hisob-kitob qilinadi
6	AttendanceRecord	Ma'lumotlar bazasiga saqlandi
7	Notifications	Ota-onaga Telegram xabar yuboriladi
8	Dashboard update	WebSocket orqali real-time yangilanish

4. FOYDALANUVCHI ROLLARI VA VAKOLATLARI

Rol	Kirish doirasi	Asosiy vakolatlar
Super Admin	Butun tizim	Barcha ma'lumotlar, rol tayinlash, tizim sozlamalari, audit
Viloyat Direktori	O'z viloyati	Viloyat statistikasi, tumanlar va maktablar nazorati
Tuman Direktori	O'z tumani	Tuman tahlili, maktablar nazorati
Maktab Direktori	O'z maktabi	Sinflar, o'quvchilar, qurilmalar, hisobotlar
Operator/IT	O'z maktabi	O'quvchi ro'yxatdan o'tkazish, qurilma boshqaruvi, ota-ona ulanishi
O'qituvchi	O'z sinflar	O'z sinfi davomadini ko'rish
Ota-ona	Faqat o'z farzandlari	Xabarnomalar, davomad tarixi

4.1 Muhim qoida: Ko'p rol

Bir foydalanuvchi bir vaqtda bir nechta rolga ega bo'lishi mumkin. Masalan, maktab direktori ayni vaqtda o'z farzandi uchun ota-ona bo'lishi mumkin. Har bir rol ma'lum bir tashkilot doirasiga (viloyat, tuman, maktab) bog'langan.

5. MA'LUMOTLAR BAZASI SXEMASI

accounts.User

Barcha foydalanuvchilar (telefon, parol, JWT auth)

```
id (UUID, PK) | phone (unique) | first_name, last_name, middle_name | is_active, is_staff, is_superuser | created_at, updated_at
```

accounts.Role

Rol ta'riflari

```
name (choices: superadmin, region_director, ...) | description
```

accounts.UserRole

Foydalanuvchi-rol bog'liq (scope bilan)

```
user (FK→User) | role (FK→Role) | region (FK→Region, null) | district (FK→District, null) | school (FK→School, null) | is_active | assigned_by (FK→User)
```

organizations.Region

Viloyatlar

```
name | code (unique) | is_active
```

organizations.District

Tumanlar

```
region (FK→Region) | name | code | is_active
```

organizations.School

Maktablar

```
district (FK→District) | name | school_number | address, phone, email | director_name | is_active
```

organizations.Class

Sinflar

```
school (FK→School) | name | grade (1-11) | section (A, B, C) | academic_year | teacher (FK→Teacher, null) | is_active
```

students.Student

O'quvchilar

```
id (UUID, PK) | school (FK→School) | class_ref (FK→Class) | first_name, last_name, middle_name | birth_date | gender (M/F) | student_id (unique) | photo | is_active
```

students.Teacher

O'qituvchilar

```
id (UUID, PK) | user (FK→User, null) | school (FK→School) | first_name, last_name | subject | employee_id (unique)
```

students.Parent

Ota-onalar

```
id (UUID, PK) | user (FK→User, null) | phone (unique) | telegram_id (unique, null) | is_registered
```

students.ParentStudentLink

Ota-ona ↔ O'quvchi bog'liq

```
parent (FK→Parent, null - pending holat) | student (FK→Student) | parent_phone |  
relationship (father/mother/guardian) | status (pending/active/rejected) | activated_at
```

devices.Device

Face ID qurilmalar

```
school (FK→School) | name | brand (hikvision/dahua/zkteco) | serial_number (unique) |  
ip_address | port | username, password | location | status | last_seen
```

devices.DeviceSyncLog

Sinxronizatsiya loglari

```
device (FK→Device) | sync_type (push/pull) | status | total_records, synced_records,  
failed_records | error_message | started_at, completed_at
```

devices.DeviceRawLog

Qurilmadan xom loglar

```
device (FK→Device) | employee_no | event_time | event_type | raw_data (JSON) | processed |  
UNIQUE: (device, employee_no, event_time)
```

attendance.AttendanceRecord

Qayta ishlangan davomad

```
student (FK→Student) | date | check_in (DateTime) | check_out (DateTime) | check_in_device  
(FK→Device) | check_out_device (FK→Device) | status (present/late/absent/excused) |  
late_minutes | raw_log (FK→DeviceRawLog) | UNIQUE: (student, date)
```

notifications.TelegramAccount

Telegram akkountlar

```
user (FK→User) | telegram_id (unique) | username | phone | is_active
```

notifications.Notification

Xabarnomalar

```
recipient (FK→User) | telegram_id | notification_type | title, message | data (JSON) |  
status | is_read
```

accounts.AuditLog

Audit loglari

```
user (FK→User) | action (create/update/delete/login) | model_name | object_id | changes  
(JSON) | ip_address | created_at
```

6. API ENDPOINTLARI

Base URL: /api/v1/

Autentifikatsiya

Metod	Endpoint	Tavsif	Ruxsat
POST	/auth/login/	Kirish → {access, refresh, user}	Hammaga
POST	/auth/refresh/	Token yangilash	Hammaga
POST	/auth/logout/	Chiqish (token blacklist)	Kirgan
GET/PATCH	/auth/me/	Profil ko'rish/tahrirlash	Kirgan
POST	/auth/change-password/	Parol o'zgartirish	Kirgan

Foydalanuvchilar

Metod	Endpoint	Tavsif	Ruxsat
GET/POST	/auth/users/	Barcha foydalanuvchilar	Operator+
GET/PATCH	/auth/users/{id}/	Foydalanuvchi batafsil	Operator+
GET/POST	/auth/users/{id}/roles/	Rollarni boshqarish	SuperAdmin
POST	/auth/users/{id}/deactivate/	Deaktivlash	SuperAdmin
GET	/auth/audit-logs/	Audit loglari	SuperAdmin

Tashkilotlar

Metod	Endpoint	Tavsif	Ruxsat
GET/POST	/regions/	Viloyatlar CRUD	SuperAdmin
GET/POST	/districts/	Tumanlar CRUD	SuperAdmin
GET/POST	/schools/	Maktablar CRUD	Direktor+
GET/POST	/classes/	Sinflar CRUD	Operator+

O'quvchilar

Metod	Endpoint	Tavsif	Ruxsat
GET/POST	/students/	O'quvchilar ro'yxati/yaratish	Operator+
GET/PATCH/DELETE	/students/{id}/	O'quvchi batafsil	Operator+
GET	/students/{id}/attendance/	O'quvchi davomadi	O'qituvchi+
GET/POST	/teachers/	O'qituvchilar CRUD	Operator+
GET/POST	/parents/	Ota-onalar ro'yxati	Operator+
GET/POST	/parent-links/	Ota-ona bog'liq	Operator+

Qurilmalar

Metod	Endpoint	Tavsif	Ruxsat
GET/POST	/devices/	Qurilmalar CRUD	Operator+

GET	/devices/{id}/status/	Qurilma holati	Operator+
POST	/devices/{id}/sync/	O'quvchilarni qurilmaga yuklash	Operator+
POST	/devices/{id}/pull-logs/	Qurilmadan log olish	Operator+
GET	/devices/{id}/sync-logs/	Sinxronizatsiya loglari	Operator+
GET	/raw-logs/	Xom loglar	Direktor+

Davomad

Metod	Endpoint	Tavsif	Ruxsat
GET	/attendance/	Davomad yozuvlari (filter bilan)	O'qituvchi+
GET	/attendance/today/	Bugungi davomad	O'qituvchi+
GET	/attendance/statistics/	Statistika	O'qituvchi+
PATCH	/attendance/{id}/	Davomadni tahrirlash	Operator+

Hisobotlar

Metod	Endpoint	Tavsif	Ruxsat
GET	/reports/daily/	Kunlik hisobot (?date=YYYY-MM-DD)	Hammaga
GET	/reports/weekly/	Haftalik hisobot	Hammaga
GET	/reports/monthly/	Oylik hisobot (?year=&month;=)	Hammaga
GET	/reports/student/{id}/	O'quvchi tarixi	Hammaga
GET	/reports/overview/	Dashboard umumiy statistika	Hammaga

Telegram Bot API

Metod	Endpoint	Tavsif	Ruxsat
POST	/telegram/link-parent/	Ota-onani bog'lash (bot dan)	Bot Secret
GET	/telegram/my-children/	Farzandlar ro'yxati (bot dan)	Bot Secret

7. FACE ID QURILMA INTEGRATSIYASI

7.1 Adapter pattern arxitekturası

Har bir Face ID qurilma brendi uchun alohida adapter sinfi yaratiladi. Barcha adapterlar BaseDeviceAdapter abstrakt sinfidan meros oladi. Yangi brend qo'shish uchun faqat yangi adapter sinfi yaratib, factory.py ga ro'yxatdan o'tkazish kifoya.

7.2 Qo'llab-quvvatlanadigan brendlar

Brend	Protokol	Autentifikatsiya	Holat
Hikvision	ISAPI REST	HTTP Digest Auth	■ Tayyor
Dahua	HTTP/CGI	HTTP Digest Auth	■ Tayyor
ZKTeco	HTTP REST	Cookie/Token	■ Tayyor
Anviz	HTTP REST	Cookie/Token	■ Tayyor
Boshqa	Sozlanuvchi	Sozlanuvchi	■ Kelajak

7.3 Adapter metodlari

- `connect()` — Qurilmaga ulanish (HTTP/TCP)
- `get_device_info()` — Model, firmware, serial raqam
- `get_users()` — Qurilmadagi barcha foydalanuvchilar
- `add_user(data)` — Yangi foydalanuvchi qo'shish
- `delete_user(id)` — Foydalanuvchi o'chirish
- `upload_face(id, image)` — Yuz rasmini yuklash
- `get_raw_logs(start, end)` — Ma'lum vaqt oralig'idagi loglar
- `get_status()` — Online/offline holat tekshirish

7.4 Sinxronizatsiya jarayoni

Push (O'quvchilarni qurilmaga yuklash)	Pull (Qurilmadan log olish)
1. Operator «Yuklash» tugmasini bosadi 2. Celery task yaratiladi 3. Qurilmaga ulaniladi 4. Barcha aktiv o'quvchilar iteratsiya qilinadi 5. Har bir o'quvchi uchun <code>add_user()</code> + <code>upload_face()</code> 6. Natija DeviceSyncLog da saqlanadi 7. WebSocket orqali UI yangilanadi	1. Operator «Log olish» tugmasini bosadi 2. Celery task yaratiladi 3. Qurilmaga ulaniladi 4. <code>get_raw_logs(last_24h)</code> chaqiriladi 5. Har bir log DeviceRawLog ga saqlanadi (takror yo'q) 6. <code>process_raw_logs</code> task ishga tushadi 7. AttendanceRecord yaratiladi 8. Ota-onaga Telegram xabar

8. TELEGRAM BOTLAR

8.1 Ota-ona boti

Ota-onalar uchun Telegram boti. O'quvchi maktabga kelganda yoki ketganda real vaqtda xabar yuboradi.

- **/start** — Ro'yxatdan o'tish (telefon raqami orqali avtomatik bog'lanish)
- **Farzandlarim** — Barcha farzandlar ro'yxati va bugungi holat
- **Bugungi davomad** — Bugungi davomad statistikasi
- **Davomad tarixi** — Oxirgi 30 kunlik tarix
- **Yordam** — Bot buyruqlari

8.2 Ota-ona auto-link jarayoni

1. Operator o'quvchi yaratishda ota-ona telefon raqamini kiritadi → ParentStudentLink (status=pending) yaratiladi
2. Ota-ona Telegram botga /start bosadi
3. Telefon raqamini ulashadi
4. Bot backend /telegram/link-parent/ ga yuboradi
5. Backend pending linkni active qiladi
6. Xabarnomalar yoqiladi ■

8.3 Boshqaruv boti

Maktab rahbarlari va operatorlar uchun. Tezkor statistika, qurilma holati va kunlik hisobotlar.

- **Bugungi hisobot** — Bugungi present/late/absent statistikasi
- **Qurilmalar** — Online/offline qurilmalar holati
- **Haftalik trend** — 7 kunlik davomad grafikasi
- **Ogohlantirishlar** — Muhim tizim ogohlantirishlari

9. WEBSOCKET (REAL-TIME)

Endpoint	Maqsad	Yuboriluvchi eventlar
ws://server/ws/attendance/	Real-time davomad feed	attendance_update
ws://server/ws/devices/	Qurilma sinxronizatsiya holati	device_event, sync_started, sync_completed

10. XAVESIZLIK TALABLARI

JWT Autentifikatsiya	Access token (60 daqiqa) + Refresh token (7 kun). Chiqishda blacklist.
Rol-asosli kirish	Har bir endpoint uchun minimal zarur ruxsat. Boshqa ma'lumotlar ko'rinmaydi.
Audit logging	Barcha muhim amallar (create/update/delete/login) AuditLog da qayd etiladi.
Bot xavfsizligi	Bot→Backend so'rovlar X-Bot-Secret header orqali tekshiriladi.
Qurilma xavfsizligi	Qurilma parollari shifrlangan. VPN yoki IP allowlist tavsiya etiladi.
Ma'lumot izolyatsiyasi	Har bir foydalanuvchi faqat o'z doirasidagi ma'lumotlarni ko'radi.
Parol validatsiya	Minimal 8 belgi. Kuchli parol tavsiya etiladi.

11. DEPLOYMENT VA INERA

11.1 Docker Compose xizmatlar

Xizmat	Port	Tavsif
postgres	5432	PostgreSQL 15 - asosiy ma'lumotlar bazasi
redis	6379	Cache va Celery broker
backend	8000	Daphne (ASGI) - HTTP + WebSocket
celery	-	Celery worker (async tasks)
celery-beat	-	Celery beat (scheduled tasks)
parent-bot	-	Ota-ona Telegram boti
management-bot	-	Boshqaruv Telegram boti
frontend	80	Nginx - React production build

11.2 Tez ishga tushirish

```
# 1. Loyihani klonlash
git clone && cd davomad_bot

# 2. Muhit o'zgaruvchilarini sozlash
cp .env.example .env
# .env faylini tahrirlang (DB, Telegram tokens, SECRET_KEY)

# 3. Docker bilan ishga tushirish
docker-compose up -d

# 4. Superadmin yaratish (avtomatik ham qilinadi)
make superadmin

# 5. Kirish: http://localhost
# Login: +998901234567 | Parol: Admin1234!
# API docs: http://localhost:8000/api/docs/
```

12. RIVOJLANISH BOSQICHLARI

1-bosqich	Asosiy tuzilma	• Loyiha bootstrap, Docker muhit • Django sozlamalari va barcha applar • Autentifikatsiya (JWT) va rol tizimi • React frontend bazasi
2-bosqich	Tashkilot tuzilmasi	• Viloyat, tuman, maktab, sinf CRUD • O'quvchi va o'qituvchi modullari • Ota-ona bog'liq tizimi
3-bosqich	Qurilma integratsiyasi	• Device modellari va adapterlar • Hikvision, Dahua, ZKTeco adapterlari • Push/Pull sinxronizatsiya • Yuz rasm yuklash
4-bosqich	Davomad mexanizmi	• Xom log qayta ishlash • Deduplication va student mapping • AttendanceRecord yaratish • Kechikish hisob-kitob
5-bosqich	Telegram botlar	• Ota-ona boti • Boshqaruv boti • Auto-link tizimi • Xabarnomalar (Celery tasks)
6-bosqich	Dashboard va hisobotlar	• Rol-asosli dashboardlar • Grafik va analitika (Recharts) • Kunlik/haftalik/oylik hisobotlar • WebSocket real-time yangilanish
7-bosqich	Xavfsizlik va polishing	• Audit loglari • Performance optimizatsiya • Monitoring va alertlar • Production deployment